One model, many switches

Measuring the effect of biological realism like for like

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Why a second vignette

The first vignette built up realism along a ladder in three host–pathogen systems. Those systems start from three different published models, so their steps cannot be compared directly, and a ladder adds things in one order only.

This vignette uses a single nested model in which each piece of biology is a switch. With every switch off it is the simplest possible model: susceptible and infectious adults. Any combination of switches can be turned on, so that:

- every model is compared with every other like for like;
- the effect of one piece of realism can be measured against every setting of the others;
- the same comparison can be repeated in different conditions.

Switch	What it adds
`latent`	infected animals are not infectious at once
`environment`	half of transmission goes through a pool of zoospores that outlasts the animals that shed them (total transmission unchanged)

Switch	What it adds
`load`	infections start light (little shedding, no disease death) and progress to heavy
`recovery`	infected animals can clear the infection
`larvae`	an aquatic larval stage between birth and adulthood
`larval_infection`	larvae are infected by adults, without disease, and half carry the infection through metamorphosis

> **!** Important
>
> The model is anchored on the Bsal model of Schmidt et al. (2017) for fire salamanders: with only `latent` on it is their published model, with their parameter values. The carry-over through metamorphosis is from Wang et al. (2021), read from the package's compendium. Everything else is new work in this package and is not a published result, and the values attached to the other switches (pool half-life 30 days, light infections shedding 10% as much and lasting about two weeks, recovery at the same rate as progression, a four-month larval stage) are illustrative.

```
using Chytridiomycosis, Plots, Printf
default(linewidth = 2, size = (720, 380), legendfontsize = 8, guidefontsize
 ↪  = 9,
    titlefontsize = 10, left_margin = 4Plots.mm, bottom_margin = 4Plots.mm)

T(; kw...) = MicrobiomeTreatment(; kw...)
bio(; kw...) = ChytridBiology(:wang2021; kw...)
schmidt = SchmidtParameters(:schmidt2017)
```

```
SchmidtParameters{Float64}(0.39, 0.125, 52.0, 364.0, 52.0, 25.0, 0.0625, 1.0)
```

The outcomes

Bsal arrives in a population at carrying capacity. Four things are recorded for each model:

- the pathogen's initial growth rate;
- untreated: mean adults over the next ten years, as a fraction of carrying capacity;
- block: the same, with every adult treated once with bacteria that block 99% of infections for life;
- survive: the same, with bacteria that prevent 99% of disease deaths and do not block infection. This is the kind of protection seen in trials.

```
function outcomes(p, b)
    run(tr; kw...) = solve(NestedChytridProblem(p, b, tr; kw...))
    u = run(T())
    return (growth = invasion_growth_rate(p, b), untreated = u.mean_adults,
     ↪  lowest = u.min_adults,
        block = run(T(efficacy = 0.99); coverage = 1.0).mean_adults,
        survive = run(T(tolerance = 0.99); coverage = 1.0).mean_adults)
end
```

```
outcomes (generic function with 1 method)
```

A check first: with only the latent class switched on, the growth rate equals the one that follows from the equations of Schmidt et al. (2017).

```
p = schmidt
λ = (sqrt((p.e - p.d_I)^2 + 4 * p.e * p.β * p.K) - 2p.b - p.d_I - p.e) / 2
(nested = invasion_growth_rate(p, bio(latent = true)), schmidt = λ)
```

```
(nested = 496.972318234428, schmidt = 496.97231824502785)
```

A ladder in one model

The switches are turned on one after another, starting from the simplest model.

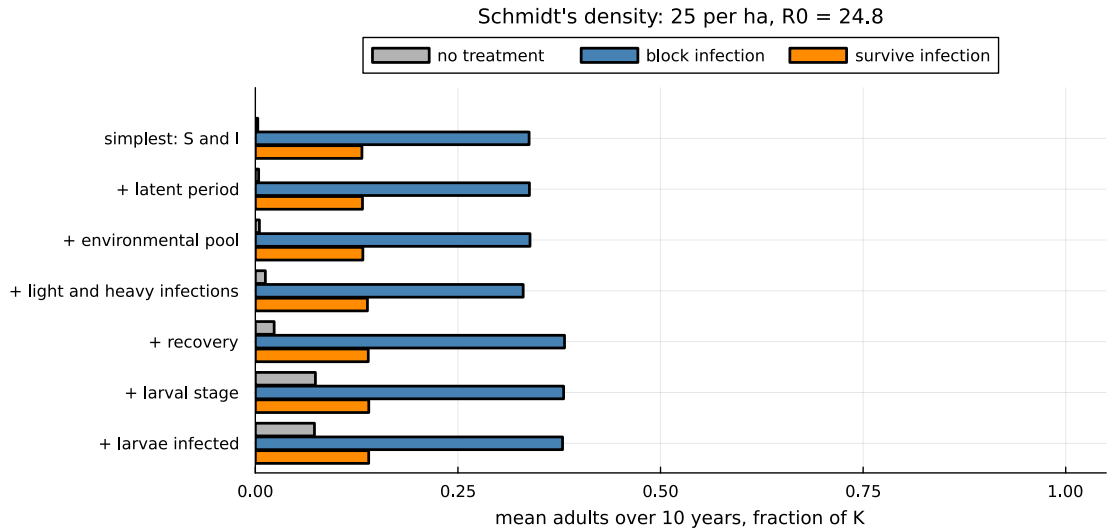
```
order = [:latent, :environment, :load, :recovery, :larvae,
 ↪  :larval_infection]
names_ = ["simplest: S and I", "+ latent period", "+ environmental pool", "+
 ↪  light and heavy infections",
    "+ recovery", "+ larval stage", "+ larvae infected"]
ladder_models(p) = [outcomes(p, bio(; (k => true for k in order[1:i])...))
 ↪  for i in 0:length(order)]

function ladder_figure(rows, title)
    k = length(rows)
    bar((1:k) .- 0.27, [r.untreated for r in rows]; orientation = :h,
     ↪  bar_width = 0.25, label = "no treatment",
        color = :gray70, yticks = (1:k, names_), yflip = true, xlims = (0,
 ↪  1.05), xticks = 0:0.25:1,
        xlabel = "mean adults over 10 years, fraction of K", size = (840, 80
 ↪  + 44k),
        left_margin = 6Plots.mm, legend = :outertop, legend_column = 3,
 ↪  title)
```

```
    bar!(1:k, [r.block for r in rows]; orientation = :h, bar_width = 0.25,
     ↪  label = "block infection", color = :steelblue)
    bar!((1:k) .+ 0.27, [r.survive for r in rows]; orientation = :h,
     ↪  bar_width = 0.25, label = "survive infection",
        color = :darkorange)
end

function ladder_table(rows)
    println("| Model | Growth rate (per yr) | Untreated | Lowest untreated |
     ↪  Block | Survive |")
    println("|---|---|---|---|---|---|")
    for (n, r) in zip(names_, rows)
        @printf("| %s | %.0f | %.3f | %.0e | %.3f | %.3f |\n", n, r.growth,
         ↪  r.untreated, r.lowest, r.block, r.survive)
    end
end

high = ladder_models(schmidt)
ladder_figure(high, "Schmidt's density: 25 per ha, R0 = 24.8")
```




```
ladder_table(high)
```

Model	Growth rate (per yr)	Untreated	Lowest untreated	Block	Survive
simplest: S and I	1248	0.003	2e-11	0.338	0.132

Model	Growth rate (per yr)	Untreated	Lowest untreated	Block	Survive
+ latent period	497	0.004	2e-11	0.338	0.132
+ environmental pool	309	0.005	2e-11	0.339	0.133
+ light and heavy infections	111	0.012	6e-06	0.331	0.138
+ recovery	98	0.023	2e-03	0.382	0.139
+ larval stage	98	0.074	2e-02	0.380	0.140
+ larvae infected	98	0.073	2e-02	0.379	0.140

At the density Schmidt et al. used, where Bsal's reproduction number is about 25, the added biology changes the conclusions very little. The epidemic slows more than ten-fold as switches are turned on, but the population collapses in every model, and each treatment gives almost the same result throughout. One thing does change: the lowest density the untreated population reaches rises by many orders of magnitude, which matters for whether it could go extinct.

Every combination

A ladder is one path. The four switches for the adult stage are now set in every combination, with three settings for the larvae (none, present but not infected, infected): 48 models.

```
configs = vec([(; latent, environment, load, recovery, larvae = lv > 0,
 ↪  larval_infection = lv > 1)
    for latent in (false, true), environment in (false, true), load in
 ↪  (false, true),
        recovery in (false, true), lv in 0:2])
all_models(p) = [(c, outcomes(p, bio(; c...))) for c in configs]

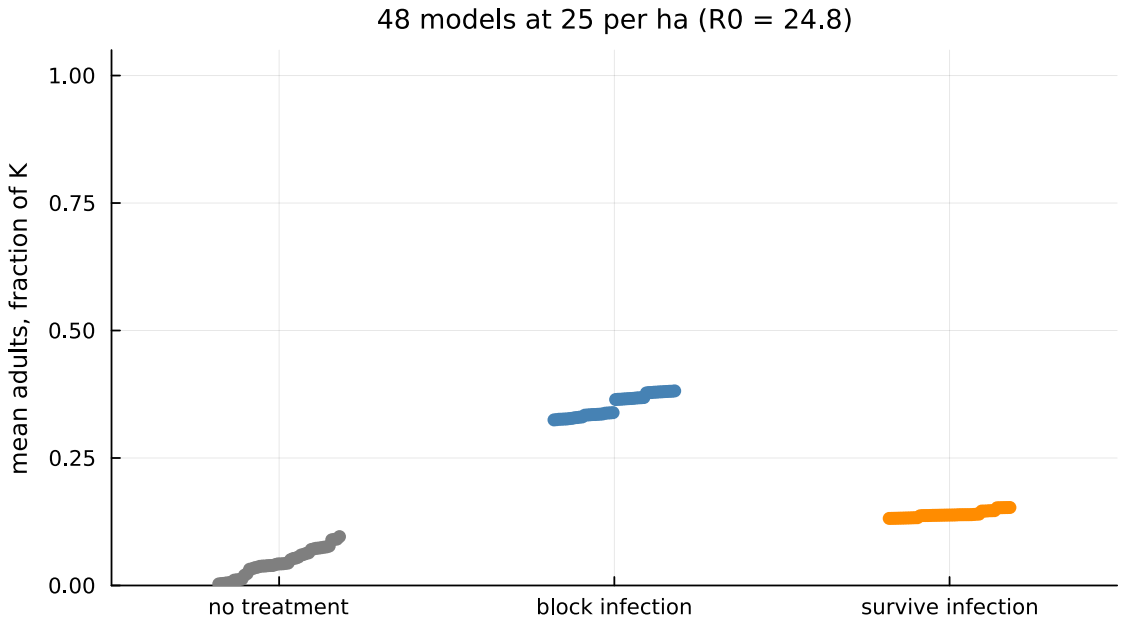
same(c, c2, skip) = all(getfield(c, k) == getfield(c2, k) for k in keys(c)
 ↪  if !(k in skip))
pieces = [
    ("latent period", c -> c.latent, (c, c2) -> !c2.latent && same(c, c2,
 ↪  (:latent,))),
    ("environmental pool", c -> c.environment, (c, c2) -> !c2.environment &&
 ↪  same(c, c2, (:environment,))),
    ("light and heavy infections", c -> c.load, (c, c2) -> !c2.load &&
 ↪  same(c, c2, (:load,))),
    ("recovery", c -> c.recovery, (c, c2) -> !c2.recovery && same(c, c2,
 ↪  (:recovery,))),
```

```
    ("larval stage", c -> c.larvae && !c.larval_infection,
        (c, c2) -> !c2.larvae && same(c, c2, (:larvae, :larval_infection))),
    ("larvae infected", c -> c.larval_infection,
        (c, c2) -> c2.larvae && !c2.larval_infection && same(c, c2,
 ↪  (:larvae, :larval_infection))),
]
# change in an outcome when a piece is switched on, for every setting of the
 ↪  other pieces
effects(res, on, off, f) = [f(o) - f(o2) for (c, o) in res if on(c) for (c2,
 ↪  o2) in res if off(c, c2)]

function effects_table(res)
    println("| Piece switched on | Untreated: mean (range) | Block: mean
     ↪  (range) | Survive: mean (range) |")
    println("|---|---|---|---|")
    for (label, on, off) in pieces
        cells = map((o -> o.untreated, o -> o.block, o -> o.survive)) do f
            d = effects(res, on, off, f)
            @sprintf("%+.2f (%+.2f to %+.2f)", sum(d) / length(d),
             ↪  minimum(d), maximum(d))
        end
        println("| ", label, " | ", join(cells, " | "), " |")
    end
end

function spread_figure(res, title)
    f = (o -> o.untreated, o -> o.block, o -> o.survive)
    labels = ["no treatment", "block infection", "survive infection"]
    plot(; xticks = (1:3, labels), xlims = (0.5, 3.5), ylims = (0, 1.05),
     ↪  ylabel = "mean adults, fraction of K", legend = false,
        title, size = (640, 360))
    for (i, g) in enumerate(f)
        v = [g(o) for (_, o) in res]
        scatter!(i .+ range(-0.18, 0.18; length = length(v)), sort(v);
         ↪  markersize = 4,
            color = (:gray50, :steelblue, :darkorange)[i], markerstrokewidth
 ↪  = 0)
    end
    current()
end

res_high = all_models(schmidt)
spread_figure(res_high, "48 models at 25 per ha (R0 = 24.8)")
```




```
effects_table(res_high)
```

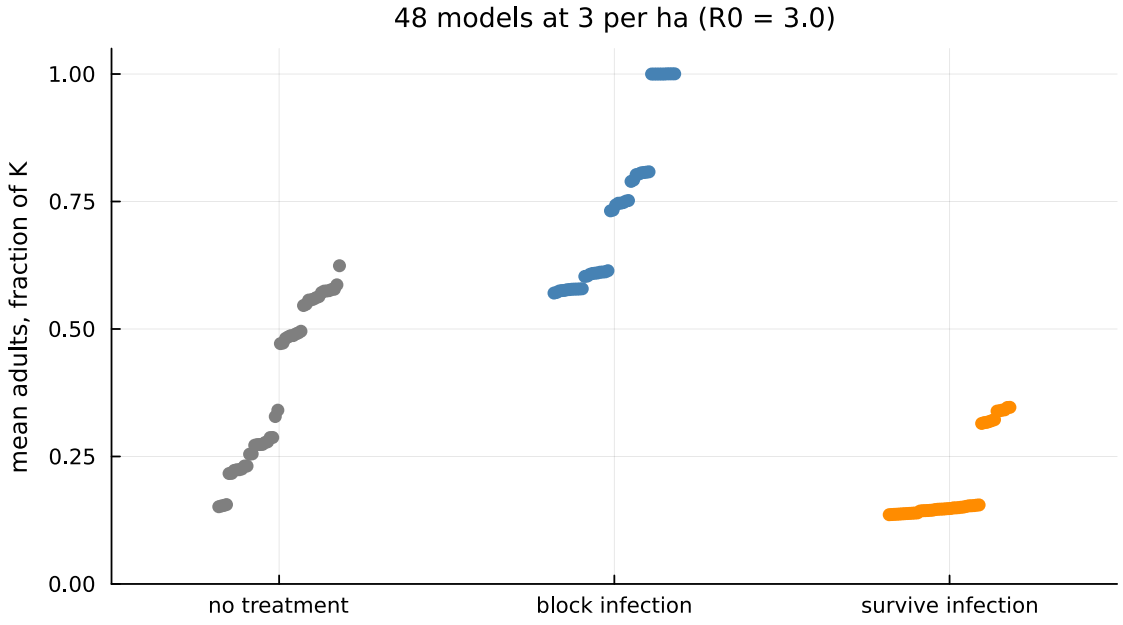
Piece switched on	Untreated: mean (range)	Block: mean (range)	Survive: mean (range)
latent period	+0.00 (+0.00 to +0.00)	+0.00 (+0.00 to +0.00)	+0.00 (+0.00 to +0.01)
environmental pool	-0.02 (-0.08 to +0.00)	+0.00 (+0.00 to +0.00)	+0.00 (-0.00 to +0.00)
light and heavy infections	+0.01 (-0.00 to +0.02)	+0.00 (-0.01 to +0.01)	-0.00 (-0.01 to +0.01)
recovery	+0.03 (+0.00 to +0.09)	+0.04 (+0.03 to +0.05)	+0.01 (+0.00 to +0.02)
larval stage	+0.02 (-0.03 to +0.05)	-0.00 (-0.00 to -0.00)	+0.00 (+0.00 to +0.00)
larvae infected	-0.00 (-0.01 to -0.00)	-0.00 (-0.00 to -0.00)	-0.00 (-0.00 to -0.00)

Each dot is one of the 48 models. At this density they sit almost on top of each other: no piece of host biology moves any outcome by more than about 0.1. When the pathogen spreads this fast and kills this quickly, the simple model gives the same answer as the complex one.

The same comparison nearer the threshold

Schmidt et al. show that Bsal cannot invade below about one adult female per hectare. The comparison is repeated at 3 per hectare (reproduction number 3) and 1.5 per hectare (1.5).

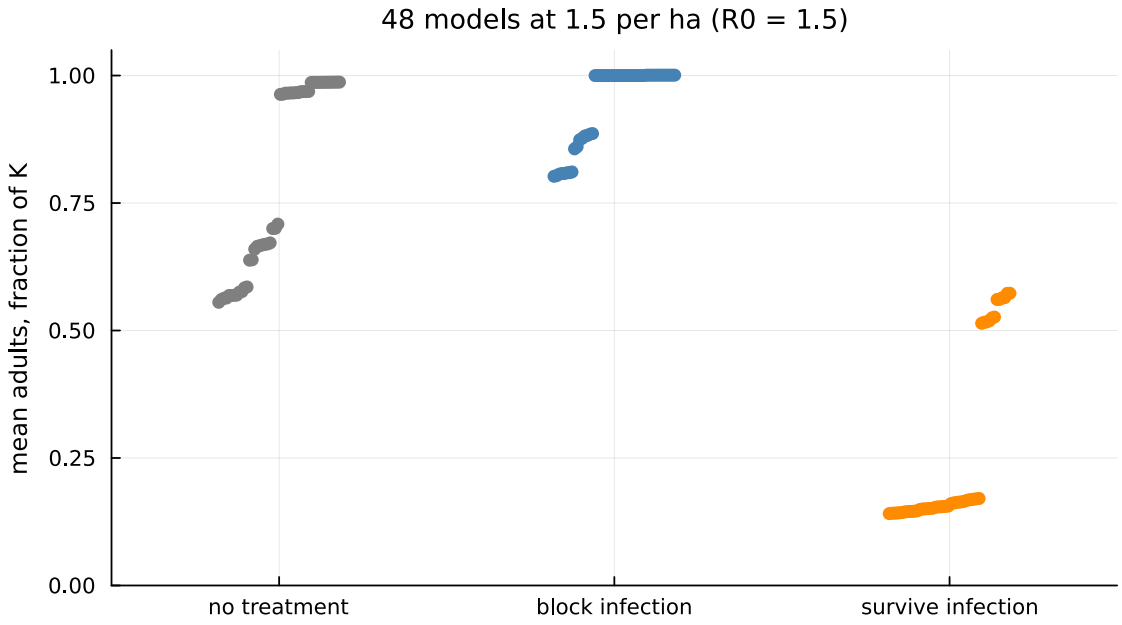
```
mid, low = Chytridiomycosis.with(schmidt; K = 3.0),
 ↪  Chytridiomycosis.with(schmidt; K = 1.5)
res_mid, res_low = all_models(mid), all_models(low)
spread_figure(res_mid, "48 models at 3 per ha (R0 = 3.0)")
```




```
effects_table(res_mid)
```

Piece switched on	Untreated: mean (range)	Block: mean (range)	Survive: mean (range)
latent period	+0.01 (-0.00 to +0.09)	+0.03 (-0.00 to +0.39)	+0.00 (+0.00 to +0.00)
environmental pool	-0.02 (-0.09 to +0.03)	-0.10 (-0.39 to +0.01)	-0.00 (-0.03 to +0.00)
light and heavy infections	+0.01 (-0.07 to +0.09)	-0.03 (-0.42 to +0.06)	-0.09 (-0.20 to +0.01)
recovery	+0.30 (+0.17 to +0.47)	+0.22 (+0.00 to +0.42)	+0.10 (+0.01 to +0.21)
larval stage	+0.04 (-0.06 to +0.13)	+0.00 (-0.00 to +0.00)	-0.00 (-0.01 to +0.00)
larvae infected	-0.03 (-0.11 to -0.00)	-0.09 (-0.40 to -0.00)	-0.00 (-0.00 to -0.00)

```
spread_figure(res_low, "48 models at 1.5 per ha (R0 = 1.5)")
```



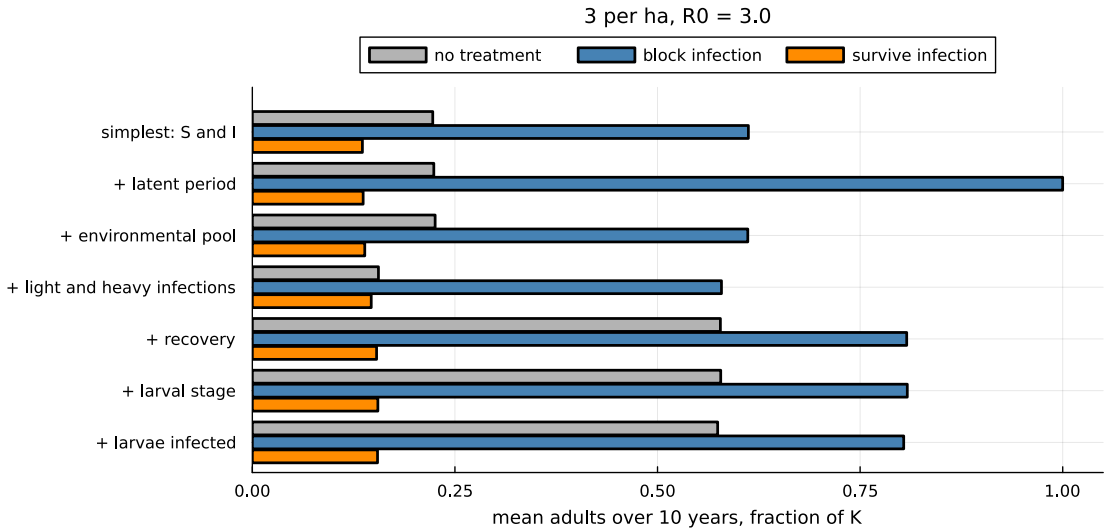

```
effects_table(res_low)
```

Piece switched on	Untreated: mean (range)	Block: mean (range)	Survive: mean (range)
latent period	+0.00 (+0.00 to +0.05)	+0.00 (-0.00 to +0.00)	+0.00 (-0.00 to +0.00)
environmental pool	-0.00 (-0.04 to +0.03)	-0.05 (-0.19 to +0.02)	-0.01 (-0.05 to +0.01)
light and heavy infections	-0.04 (-0.12 to +0.02)	-0.02 (-0.08 to -0.00)	-0.18 (-0.41 to +0.01)
recovery	+0.36 (+0.26 to +0.43)	+0.11 (+0.00 to +0.20)	+0.21 (+0.01 to +0.43)
larval stage	-0.00 (-0.04 to +0.01)	+0.00 (-0.00 to +0.00)	-0.00 (-0.01 to +0.00)
larvae infected	-0.01 (-0.07 to -0.00)	-0.04 (-0.19 to -0.00)	-0.00 (-0.00 to -0.00)

Now the models disagree. At 3 per hectare the untreated outcome ranges from 0.15 to 0.62 depending on which biology is included, and the infection-blocking treatment from 0.57 to full protection. Realism matters most near the threshold.

The ladder at 3 per hectare:

```
ladder_figure(ladder_models(mid), "3 per ha, R0 = 3.0")
```



Three things stand out.

1. Recovery is the piece that matters most, for the untreated population and for the blocking treatment. This project has no estimate of it for Bsal.
2. Pieces do not have a fixed effect. The ranges in the tables are wide: an environmental pool, or infected larvae, changes the blocking treatment by anything from nothing to −0.4 depending on what else is in the model.
3. Bacteria that let animals survive infection do worse than no treatment at these densities, in every one of the 48 models (orange below grey; counts in the table below). Untreated, an epidemic at low density burns out and the population recovers. Treated animals that survive stay infectious in this model, and keep infecting the untreated animals born after them.

```
println("| Density | Models in which the survive treatment does worse than
 ↪  none | Blocking does worse than none |")
println("|---|---|---|")
for (label, res) in (("25 per ha", res_high), ("3 per ha", res_mid), ("1.5
 ↪  per ha", res_low))
    @printf("| %s | %d of 48 | %d of 48 |\n", label, count(x -> x[2].survive
     ↪  < x[2].untreated, res),
        count(x -> x[2].block < x[2].untreated - 1e-6, res))
end
```

Density	Models in which the survive treatment does worse than none	Blocking does worse than none
25 per ha	0 of 48	0 of 48
3 per ha	48 of 48	0 of 48
1.5 per ha	48 of 48	0 of 48

Does the order of adding matter?

The effect of each piece when it is the first thing added to the simplest model, and when it is the last thing added to the otherwise complete model, at 3 per hectare:

```
full = (latent = true, environment = true, load = true, recovery = true,
 ↪  larvae = true, larval_infection = true)
empty_ = (latent = false, environment = false, load = false, recovery =
 ↪  false, larvae = false, larval_infection = false)
get_(res, c) = only(o for (c2, o) in res if c2 == c)
println("| Piece | Untreated: added first | added last | Block: added first
 ↪  | added last |")
println("|---|---|---|---|---|")
for (label, on, off) in (("latent period", (latent = true,), (latent =
 ↪  false,)),
        ("environmental pool", (environment = true,), (environment =
 ↪  false,)),
        ("light and heavy infections", (load = true,), (load = false,)),
        ("recovery", (recovery = true,), (recovery = false,)),
        ("larval stage, infected", (larvae = true, larval_infection = true),
 ↪  (larvae = false, larval_infection = false)))
    first_ = (get_(res_mid, merge(empty_, on)), get_(res_mid, empty_))
    last_ = (get_(res_mid, full), get_(res_mid, merge(full, off)))
    @printf("| %s | %+.2f | %+.2f | %+.2f | %+.2f |\n", label,
     ↪  first_[1].untreated - first_[2].untreated,
        last_[1].untreated - last_[2].untreated, first_[1].block -
 ↪  first_[2].block, last_[1].block - last_[2].block)
end
```

Piece	Untreated: added first	added last	Block: added first	added last
latent period	+0.00	+0.00	+0.39	+0.00
environmental pool	+0.00	+0.03	-0.00	+0.01
light and heavy infections	-0.07	+0.09	-0.04	+0.06
recovery	+0.33	+0.36	+0.39	+0.23
larval stage, infected	+0.05	-0.00	-0.01	-0.00

Where the two columns differ, the piece interacts with the others, and a ladder would give a different impression depending on where the piece was placed.

What this shows

1. Whether realism matters depends on the regime. For a pathogen far above its invasion threshold, six pieces of host biology barely change the outcome or the value of treatment. Near the threshold the same pieces change both a great deal.
2. So the first question is where the system sits. If Bsal in dense salamander populations really has a reproduction number near 25, a simple model is enough to say that an infection-blocking treatment given once to adults leaves about a third of the population. In sparser populations the answer depends on biology that has not been measured, recovery above all.
3. Effects of single pieces are not transferable. They vary with the rest of the model, so the result of adding one piece to a simple model should not be read as the effect of that piece.
4. Protection that keeps infected animals alive can be worse than none, where the untreated epidemic would have burnt out.

Limits

- The nested model covers the salamander system only. The toad and frog systems of the first vignette have features it lacks: a seasonal cohort, and a continuous load on each animal.
- The model is deterministic. A pathogen density can fall to a level that means less than one infected animal in a very large area and then grow back. Some differences between models near the threshold, such as full protection in one model and partial in the next, come from this, and a stochastic version is needed to confirm them.
- Treated animals that survive infection never clear it unless `recovery` is on.
- The values attached to the switches are illustrative. Different values would change the sizes of the effects, though the contrast between regimes should hold.

References

- Schmidt BR et al. (2017) R Soc Open Sci 4:160801. doi:10.1098/rsos.160801
- Wang Y et al. (2021) Nat Commun 12:5788. doi:10.1038/s41467-021-26127-9